

Computational Fluid Dynamics

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We want to simulate the flow of a non-Newtonian fluid through a two-dimensional rectangular duct $\Omega = (0, 3) \times (0, 1)$. The problem is governed by the non-Newtonian Stokes equations

$$\left\{ \begin{array}{ll} -\operatorname{div}(2\mu_{\text{eff}}\mathbf{D}) + \nabla p = \mathbf{0} & \text{in } \Omega \\ \operatorname{div} \mathbf{u} = 0 & \text{in } \Omega, \\ \mathbf{u} = (y(1-y), 0)^T & \text{on } \Gamma_1 = \{(x, y) \in \partial\Omega : x = 0\}, \\ (-p\mathbf{I} + 2\mu_{\text{eff}}\mathbf{D})\mathbf{n} = \mathbf{0} & \text{on } \Gamma_2 = \{(x, y) \in \partial\Omega : x = 3\}, \\ \mathbf{u} = \mathbf{0} & \text{on } \Gamma_{3,4} = \{(x, y) \in \partial\Omega : y = 0 \text{ or } y = 1\}, \end{array} \right. \quad (1)$$

where $\mathbf{D} = \frac{1}{2}(\nabla \mathbf{u} + \nabla^T \mathbf{u})$. The effective viscosity $\mu_{\text{eff}} = \mu_{\text{eff}}(\dot{\gamma})$ depends on the shear-rate $\dot{\gamma} = \sqrt{\mathbf{D} : \mathbf{D}}$, through the so-called power-law

$$\mu_{\text{eff}}(\dot{\gamma}) = \mu_0 \dot{\gamma}^{n-1}$$

for given constants $\mu_0 > 0$ and $n \geq 1$.

- Introduce a suitable linearization of problem (1) and the weak formulation of the linear problem to be solved at each fixed-point iteration.
- Starting from the template `template130624.ipynb` which solves the Stokes equations, write a **Firedrake** notebook for the solution of problem (1) using $\mathbb{P}_2 - \mathbb{P}_1$ finite elements for velocity and pressure, respectively. As an initial guess of the fixed-point algorithm, consider the Stokes solution with $\mu = \mu_0$. Define and use a suitable convergence criterion for the nonlinear iterations.
- Solve the nonlinear problem for $\mu_0 = 1$ and different values of $n = 1, 0.6, 0.2$, reporting the number of nonlinear iterations required to reach convergence for each value of n .
- Postprocess the final result in Paraview for the three case $n = 1, 0.6, 0.2$, saving the velocity plots (arrows colored by velocity magnitude) as `SURNAME_NAME.n.png`.
- Compute and report the x component of the force exerted by the flow on the duct walls Γ_3 and Γ_4 for the different values of n .
- Comment the results.